

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	LTVIP2026TMIDS89849
Project Name	dog breed identification using transfer learning
Maximum Marks	4 Marks

Technical Architecture:

The proposed system is a web-based application designed to identify dog breeds using transfer learning. The architecture consists of multiple layers to ensure scalability, accuracy, and efficient performance.

The **User Interface layer** allows users to upload dog images and view prediction results through a simple and responsive web application developed using HTML, CSS, and JavaScript.

The **Application Logic layer** handles user authentication, image upload, and communication between the frontend and backend. The backend is developed using Python and the Flask framework.

The **Data Preprocessing layer** processes uploaded images by resizing and normalizing them to match the model input requirements. Libraries such as OpenCV and NumPy are used in this stage.

The **Machine Learning layer** uses pre-trained deep learning models such as MobileNet or ResNet. Transfer learning improves accuracy and reduces training time by using previously learned image features.

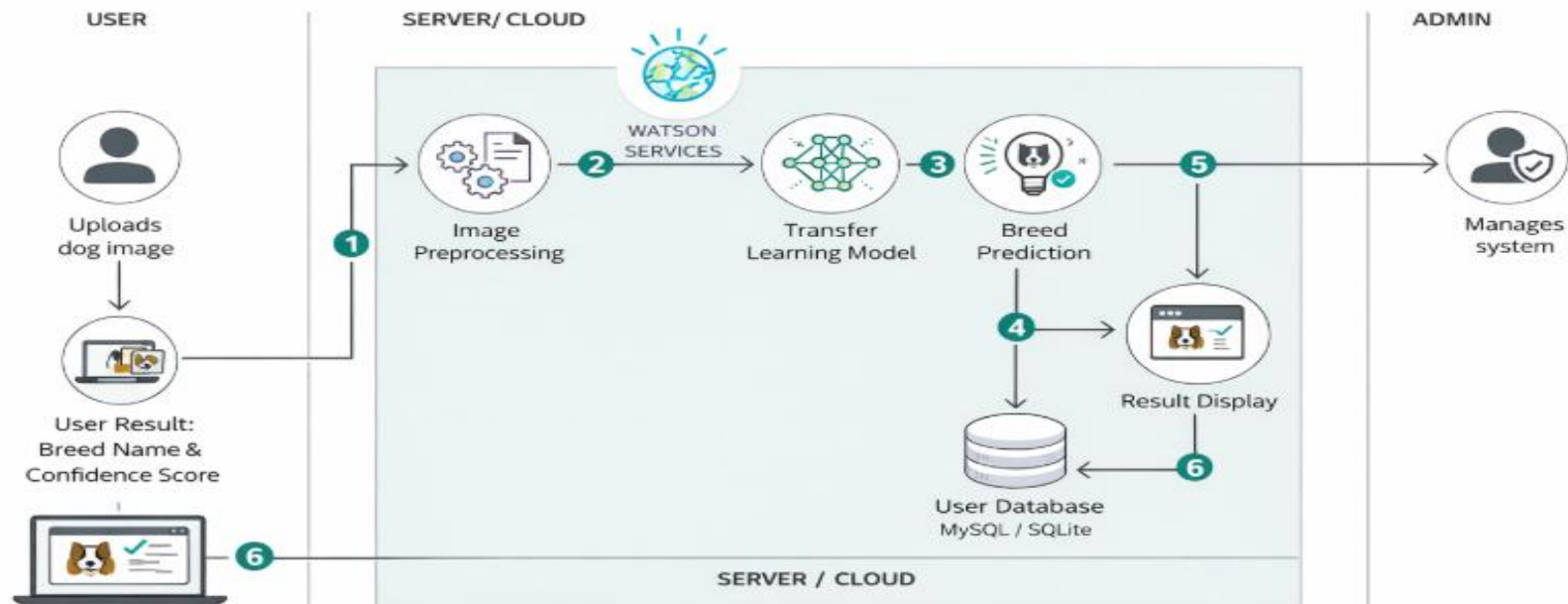


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handles user authentication, request processing, and image upload.	Python, Flask
3.	Application Logic-2	Performs image preprocessing such as resizing, normalization, and data transformation.	Python, OpenCV, NumPy
4.	Application Logic-3	Handles classification and prediction of dog breeds.	TensorFlow, Keras, Transfer Learning
5.	Database	Stores user information, login credentials, and prediction history.	MySQL / SQLite
6.	Cloud Database	Used for large-scale deployment and storage in future.	Firebase / AWS RDS
7.	File Storage	Stores uploaded dog images temporarily before prediction.	Local File System
8.	External API-1	Used for authentication and secure login in future.	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Predicts dog breed from uploaded images using deep learning.	Pre-trained CNN (MobileNet / ResNet)